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EDUCATION

Oregon State University — **Expected Graduation: Spring 2026**
Bachelor of Science, Biology, Minor: Biochemistry/Molec. Bio. (GPA 3.37) — *Corvallis, OR*

- **Relevant Coursework:** General Chemistry, Organic Chemistry, General Biochemistry, Genetics, Evolution, Vertebrate Physiology, Cell & Molecular Biology, Microbiology, Calculus (Differential & Integral), Statistics, Python for Molecular Biologists, Computer Programming for non-CS Majors, Intro. Biological Data Science, Applied Bioinformatics, Adv. Computing for Bio. Data Analysis.

EXPERIENCE

Oregon State University — *Corvallis, OR*

Bioinformatics/Microbial Genomics Research, OSU Chang Lab — **Jan. 2025 – Present**

- Generated and managed high-quality genome assemblies for *Phytophthora* isolates, handling 100+ isolates and 300+ assemblies across primary, hap1, and hap2 haplotypes.
- Executed large-scale comparative genomics workflows using GENESPACE, nf-core/pairgenomealign, and MUMmer to perform synteny and evolutionary analyses.
- Automated HPC workflows by writing SLURM batch and array scripts, integrating AGAT, Helixer, and other genome annotation tools in Singularity containerized environments.
- Produced data visualizations in R (phylogenetic trees, synteny dotplots, repeat-density plots) for research posters, presentations, and lab meetings.
- Collaborated with postdoctoral researchers to refine workflows, troubleshoot computational challenges, and contribute reproducible pipelines for departmental projects.

Undergraduate Research, Biology — **Sep. 2023 – Dec. 2023**

- Designed and conducted an 8-week research project on nematode–vegetation relationships, analyzing population diversity and abundance under enriched vs. unenriched soil conditions.
- Applied molecular biology techniques (PCR, gel electrophoresis, microscopy) to identify nematode species and validate field observations.
- Performed quantitative data analysis using R and Excel to evaluate species diversity trends; synthesized results into a formal report and delivered an oral presentation to faculty.
- Strengthened project leadership skills by coordinating group responsibilities and experimental design

Chemistry Lab, General, Analytical, and Organic — **Sep. 2023 – Jun. 2024**

- Mastered lab tools, software (LoggerPro, Excel), and techniques (titration, use of probes & chromatographs) for chemical property analysis and analyte determination in General/Organic/Analytical Chemistry coursework.
- Developed competency in organic chemistry methods, including synthesis, structure determination (IR, NMR spectroscopy, melting point determination), prediction of theoretical & experimental reactions.

LEADERSHIP

Club Secretary, Vice-President — **Jun. 2024 – Present**
Oregon State University Integrative Biology Club — *Corvallis, OR*

- Lead operations and programming for a 500+ member student organization, collaborating with faculty and peers to expand academic, professional, and community development opportunities.
- Organized and facilitated professional workshops and graduate/undergraduate research information sessions connecting students with faculty and helping them build confidence and experience.

Biology Learning Assistant (LA) — **Sep. 2024 – Jun. 2025**
Oregon State University, College of Science — *Corvallis, OR*

- Supported the BI 22X introductory biology series by facilitating active learning, assisting in lectures, and holding weekly office hours for 1000+ students.
- Attended weekly content prep meetings to discuss content and delivery with other LAs and instructors.
- Delivered individualized tutoring and small-group support to improve student comprehension and retention of core biology concepts.

Alumni Relations Chairman — **Apr. 2025 – Feb. 2025**
Phi Kappa Psi Fraternity, Oregon Beta Chapter — *Corvallis, OR*

- Managed communications with 1,000+ alumni through bi-monthly newsletters and event updates.
- Coordinated major events, including a flagship Alumni Golf Mixer (budget $1,500; 50+ attendees) and the annual Founder's Day Banquet (budget $20,000; 300+ attendees).